Low Oil Pressure Indicating System – General Data

1. General

The oil pressure switch does a dual function, as it is integral to the optional propeller de-ice system (when aircraft is equipped) and as a redundant low oil pressure sensor. If the engine is running and there is a low pressure condition (oil pressure drops to less than 40 – 42 psi) sensed at the oil pressure switch, then this will trigger a Left/Right Oil Pressure caution CAS message. The switch will de-activate when engine oil pressure rises above 44 – 46 psi.

2. Adjustment/Test

A. Function Test

Equipment required:

- Pressure tester SD5688–3, or
- Pressure tester, Barfield Model 2311F with 0 – 160 psi test gauge, nipple AN816–4 and hose Aeroquip 601000–4–0240 and a digital multimeter.

1) Disconnect oil line to pressure switch.

2) Connect hose from tester to pressure switch.

3) Connect a multimeter to the pressure switch pins and monitor the output impedance from the unit.

4) Using handpump (Tester SD5688) or screw on (Tester Barfield) tester, apply pressure to pressure switch and check that impedance shows high (open circuit) when the applied pressure is between 40 to 45 psi.

5) Slowly decrease pressure to pressure switch by carefully opening bleed valve (Tester SD5688) or by counterclockwise rotation of screw (Tester Barfield) and check that the impedance goes low (closed circuit) when the applied pressure is at 41 ± 2 psi.

6) Disconnect tester and reconnect line to pressure switch.

7) Repeat test for low oil pressure switch on other engine.

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